

ch8 Work, Energy and Power

3. Read the following passage about braking in vehicles and answer the questions that follow.

Mechanical braking is the most common braking method in vehicles. In this method, wheels of vehicles are fitted with brake pads which apply frictional forces that inhibit the motion of the wheels. Frictional braking results in a conversion of the vehicle's kinetic energy to thermal energy which then dissipates to the atmosphere in the form of waste heat.

The thermal energy (in J) generated during braking is approximately given by :

$$\text{thermal energy generated} = F_f \times d$$

where F_f is the frictional forces (in N), and
 d is the braking distance (in m).

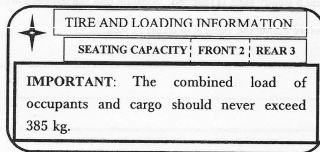
Applying the principle of energy conservation :

kinetic energy dissipated = thermal energy generated

$$\frac{1}{2}mv^2 = F_f \times d$$

From this equation it can be seen that increasing the velocity v or mass m of an object means the applied frictional forces must be increased in order to bring the object to a stop within the same distance.

- (a) Referring to the passage, explain why there should be a maximum load limit for a vehicle (see the vehicle information label below). (4 marks)



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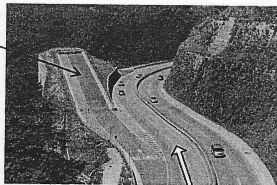
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- (b) (i) It is known that the brake pads would provide much smaller frictional forces or even brake failure would occur (i.e. without any friction provided) when reaching a temperature of over a few hundred degrees Celsius. Explain why it is not recommended for a vehicle driver to apply the brakes continuously during a long downhill trip. (1 mark)

- (ii) The figure shows an emergency escape ramp (slanting upwards) built for stopping vehicles with brake failure resulting from the situation described in (b)(i). If such a ramp makes an angle of 30° with the horizontal and a vehicle with brake failure enters the bottom end of the ramp at a speed of 25 m s^{-1} , estimate how far it will travel along the ramp before it stops. Neglect air resistance and mechanical resistances within the vehicle. ($g = 9.81 \text{ m s}^{-2}$) (2 marks)

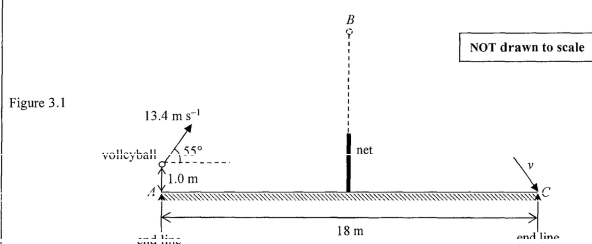
emergency escape ramp



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3. A volleyball player serves by hitting the ball from rest at a height of 1.0 m above the end line of the court. The initial speed of the ball is 13.4 m s^{-1} making an angle of 55° with the horizontal. It moves in a vertical plane perpendicular to the end line and finally reaches point C on the opposite end line as shown in Figure 3.1. Neglect the size of the ball and air resistance. ($g = 9.81 \text{ m s}^{-2}$)



- (a) (i) The mass of the volleyball is 0.22 kg. Find the work done on the ball by the player. (2 marks)

- (ii) Determine the speed v with which the ball hits point C on the ground. (2 marks)

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- (b) The length of the court AC is 18 m and the net is positioned midway between A and C. It takes time t for the ball to reach point B which is vertically above the net.

- (i) State whether the ball is ascending, flying horizontally or descending at B. (1 mark)

- (ii) Find t . (2 marks)

- (c) Another player suggests that the volleyball can reach point C in a shorter time if it is served with a similar initial speed but at a smaller angle with the horizontal (e.g. 13.2 m s^{-1} at an angle of 35°). Without doing any calculation, explain whether this suggestion is justified. (2 marks)

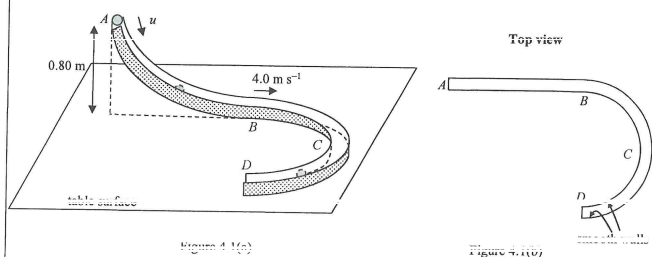
- (d) Volleyball players have to jump and land frequently in a game. Referring to principles of mechanics, explain why volleyball courts with wood rather than concrete flooring may help to protect the players from injuries. (2 marks)

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Q4

4. Figure 4.1(a) shows a smooth track $ABCD$ consisting of a sloping part AB and a horizontal semicircular part BCD fixed on a table. Point A is at 0.80 m above the table. The top view of the track is shown in Figure 4.1(b).



A small sphere is projected along the track with speed u at A . It moves along the track and passes B at 4.0 m s^{-1} . Neglect air resistance. ($g = 9.81\text{ m s}^{-2}$)

- (a) By using the law of conservation of energy, find u .

(2 marks)

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*(b) The sphere then moves along BCD with constant speed 4.0 m s^{-1} . The radius of BCD is 0.20 m .

- (i) Explain why, even though the speed of the sphere remains constant, its acceleration is not zero during the motion described above. (1 mark)

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- (ii) Estimate the magnitude of the acceleration of the sphere.

(1 mark)

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- (iii) What force provides the centripetal force for the sphere?

(1 mark)

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2.



Figure 2.1

In a road test, John drives his car along a straight horizontal road (see Figure 2.1). The car takes 9.3 s to accelerate from rest to 100 km h^{-1} . The total mass of John and his car is 1400 kg.
(Note: $100 \text{ km h}^{-1} = 27.8 \text{ m s}^{-1}$)

- (a) Find the total kinetic energy of John and his car when travelling at 100 km h^{-1} . Hence estimate the average output power of the car during this acceleration. (3 marks)

- (b) A similar road test is conducted on an inclined road. The car now takes more than 9.3 s to accelerate from rest to 100 km h^{-1} along the road (see Figure 2.2). Assume the output power of the car remains unchanged. Explain why it takes a longer time for the car to accelerate up an inclined road than along a horizontal road. (2 marks)



Figure 2.2
